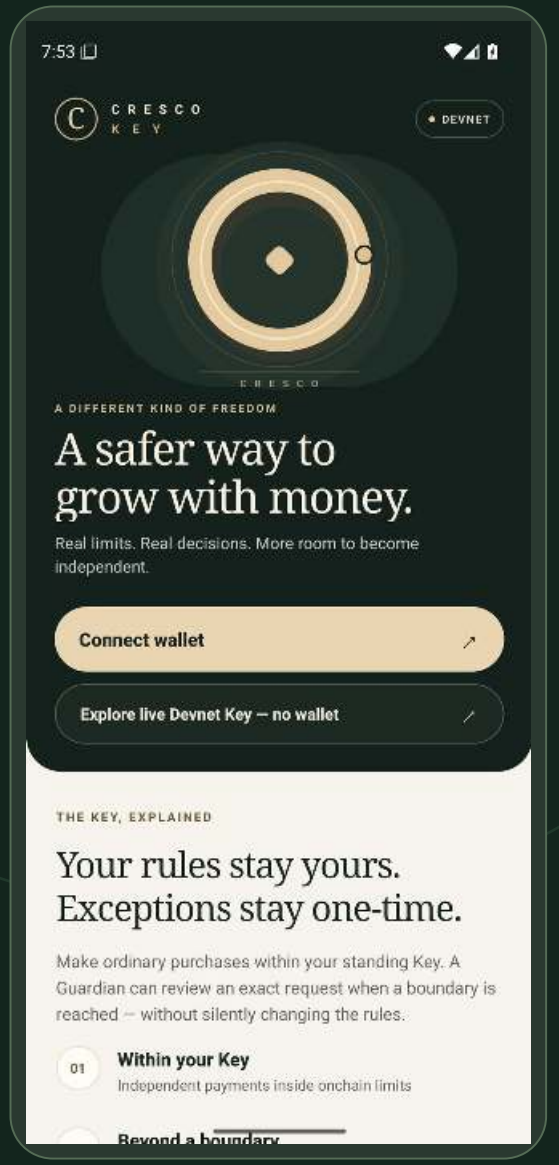
CRESCO *Key*

Freedom with boundaries.

The exception moves.
The boundary does not.

A mobile-first financial autonomy app. Real onchain limits. Precise Guardian decisions. Honest proof.



REAL ANDROID BUILD · PR #85

Independence should not mean unlimited access.

Young people need room to act. Guardians need meaningful boundaries, not permission fatigue.

OLD DEFAULT / LOCKDOWN

Ask for everything.

Every purchase becomes a parental interruption. Small decisions never become independent.

OLD DEFAULT / OPEN ENDED

Raise the limit.

A one-off need changes standing permission. The exception quietly becomes the new normal.

THE PRODUCT OPPORTUNITY

Trust the person. Keep the rule.

Explicit boundaries; precise exceptions; inspectable results.

Design problem and product hypothesis; not a measured user-survey claim.

One standing Key. One exact exception.

The two concepts are separate onchain. An allowance cannot silently rewrite the family’s permanent rule.



STANDING KEY

Rules for normal life.

Per-action cap / Period cap / Mint / Nonce

AN EXACT ONE-TIME REQUEST

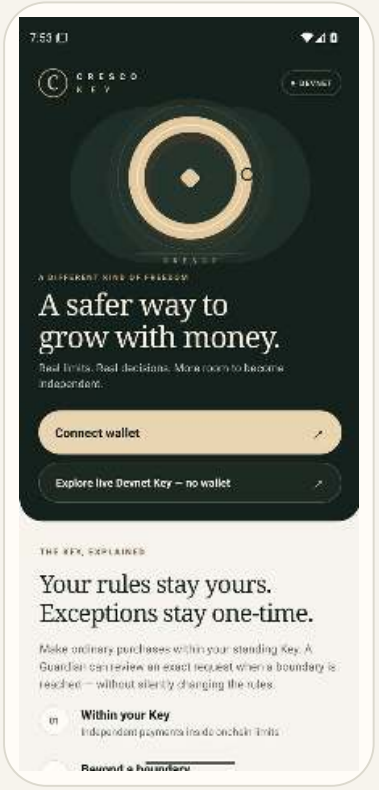
Amount + Recipient + Asset + Request ID + Nonce

GUARDIAN MAY REFUSE

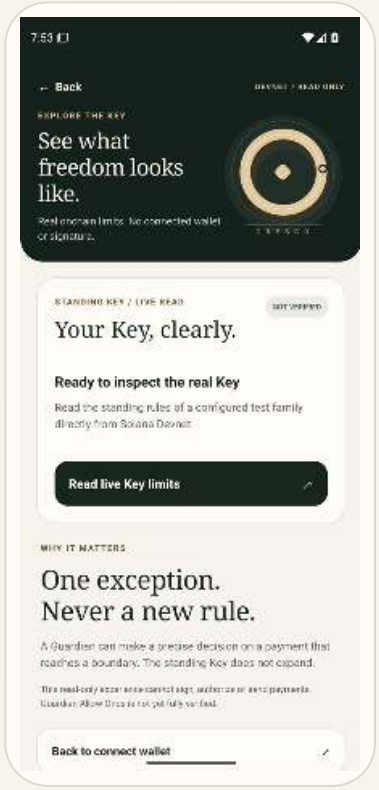
GRANT ≠ PAYMENT

Not a render. A native app.

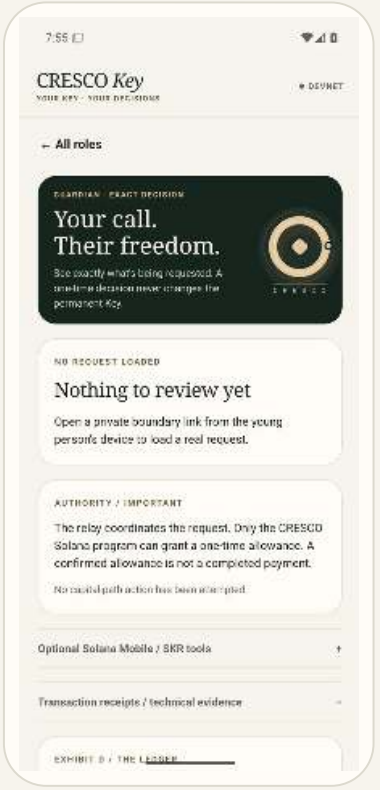
Three real paths captured from the Signature APK workflow at commit 0bd51d53.



START / WALLET



EXPLORE / RPC READ

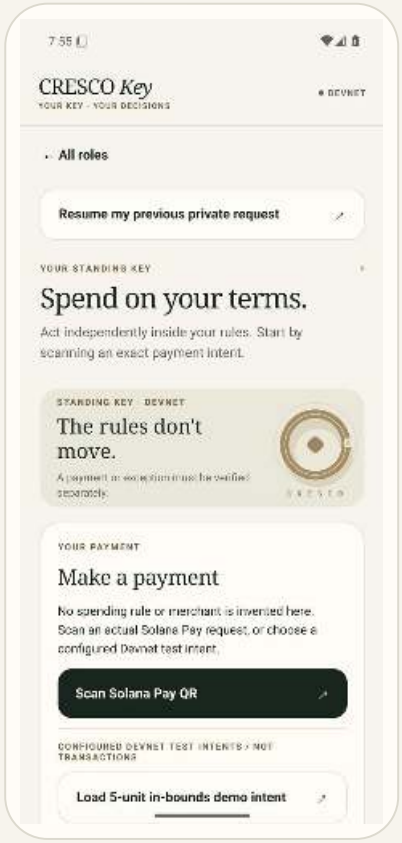


GUARDIAN / EXACT REVIEW

These are emulator captures. No fabricated balances, synthetic success states or physical-device proof.

Autonomy before approval.

A clear intent, a real standing Key and a reasoned next step when a boundary is reached.



REAL YOUNG SCREEN



CONFIGURED DEVNET INTENT

01 Inspect

Read actual onchain spending rules from the connected Key.

02 Act

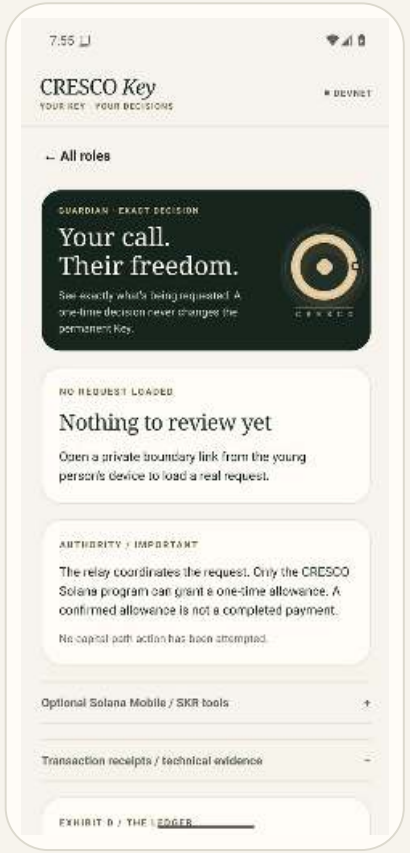
Scan Solana Pay or load the configured Devnet test intent.

03 Face the boundary

REFUSE, UNKNOWN and BLOCKED must remain distinct. No fake success.

Review the action. Not just the person.

A precise decision surface where amount, mint, recipient and nonce determine the authority requested.



NO REQUEST LOADED / REAL

EXACT ACTION / WHEN A REAL REQUEST EXISTS

ASSET	Token mint is explicit.
AMOUNT	Exact base units — no invented fiat figure.
DESTINATION	Recipient address bound to the request.
DECISION	Not this time OR Allow this action once.

Only the Solana program can grant or consume financial authority. Relay state does not transfer funds.

Real consequences. Exact receipts.

Current distinct program: 6SoGab...6DeJP. Transactions below used developer-controlled signers, not the Samsung device.

01 / PAYMENT VERIFIED

5 UNITS

Vault 100 → 95. Recipient 0 → 5. Confirmed token movement.

OPEN ON SOLANA EXPLORER ↗

02 / RULE REFUSAL VERIFIED

12 UNITS

Limit exceeded. Program rejects the attempt; no successful transfer.

OPEN ON SOLANA EXPLORER ↗

03 / GUARDIAN GRANT VERIFIED

EXACT GRANT

Onchain one-time allowance created. Redemption is still blocked.

OPEN ON SOLANA EXPLORER ↗

ONE-TIME REDEMPTION ON CURRENT PROGRAM = BLOCKED (#76)

The UI cannot outrun the chain.

A payment is successful only when the appropriate confirmation and account consequences can be established.

INSPECTED

Real data read from RPC. Not payment authority.

READY FOR ATTEMPT

A transaction may be proposed. Nothing transferred.

GRANT SUBMITTED

Permission is not the same as redemption.

REFUSED / BLOCKED

No hidden widening of the spending boundary.

UNKNOWN

Do not retry blindly or label successful.

ONCHAIN VERIFIED

A confirmed transaction plus relevant state evidence.

Explicit negative paths are a core part of the product experience, not demo errors to conceal.

Mobile is the interface. Solana is the authority.

The Guardian coordination relay cannot move capital or silently change a permanent Key.



CLOUDFLARE RELAY / coordinates exact requests, never grants spending authority

The app records source, confirmation and UNKNOWN separately. Wallet signMessage is not a payment.

Built for the device. Not a web wrapper.

Mobile product development beyond the existing CRESCO authority engine.



MOCK WALLET / MESSAGE SIGN

REACT NATIVE / EXPO

Installable standalone Android APK and native flows.

MOBILE WALLET ADAPTER

Connect, decline, authorize and sign a proof message.

SOLANA PAY / CAMERA

QR scanning and exact token payment intent parsing.

MOBILE RELAY CAPABILITIES

Private coordination tokens secured on device.

DEEP LINKS + LIFECYCLE

Resume an exact request and restore wallet/role state.

MOCK MWA EMULATOR VERIFIED / PHYSICAL WALLET PAYMENT NOT PROVEN

A reason to come back: normal spending.

CRESCO is not designed around a one-off Guardian approval demo. The normal product is recurring autonomy.

01

Most days

A young person makes routine purchases under a rule already understood.

02

Occasional exception

Only the precise out-of-bounds action prompts the Guardian.

03

Trust built over time

Families discuss limits without turning every purchase into surveillance.

PRODUCT HYPOTHESIS, NOT CLAIMED ADOPTION: no fabricated user counts, retention rates, spending volume or testimonials.

SKR opens a conversation. Never a credit line.

A separate, opt-in Solana Mobile feature. The core payment experience does not require SKR.

MAINNET / PUBLIC TOKEN ACCOUNT CHECK

Verify SKR presence.

No wallet secret. No transfer. No staking.

01 User opts in to public SKR check

02 Mainnet token presence gates prompts

03 Holder reviews a genuinely inspected Devnet Key

04 Shares a read-only family boundary record

Source-level implementation. No SKR-holder physical-device verification or bounty eligibility claimed.

Bounded authority, not approval theater.

The differentiator is enforced in the payment primitive, not just shown in a beautiful mobile interface.

SIMPLE ALLOWANCE UI	CRESCO KEY
One balance or broad spending access	**Explicit per-payment and per-period Key limits**
Parent approves a vague category	**Guardian reviews amount + mint + recipient + nonce**
Temporary need can reset a standing rule	**Exact allowance stays separate from permanent authority**
App/backend says “approved”	**The chain decides; relay only coordinates**
UI hides awkward failures	**UNKNOWN, REFUSED and BLOCKED are first-class states**

The final one-time redemption on the currently deployed program is still blocked and is not represented as proven.

A demo is not a deposit. A grant is not a payment.

Every major claim is classified by the strongest available evidence.

BUILT / LOCAL	Standalone Android APK, 5/5 CI including emulator and Mock MWA
ONCHAIN / VERIFIED	5-unit developer-signed transfer and 12-unit limit rejection
ONCHAIN / VERIFIED	Exact Guardian allowance granted; unused allowance recorded
NOT PHYSICALLY VERIFIED	Current Signature APK payment via real Solflare on a phone
BLOCKED (#76)	Exact redemption, mutation ordering and replay against current program
NOT CLAIMED	Production custody, mainnet capital, adoption or store launch

Independent evidence: GitHub Actions, Android screenshots, deployed program, Solana Explorer transactions and issue #76.

From research prototype to family-ready product.

Success is a repeatable real-user loop—not just a green CI run or a great screenshot.

01 Repair exact redemption

Reproduce #76 using the deployed binary, correct the proven root cause, verify mutation and replay.

02 Provision independent users

Authorized Young + Guardian wallets, matching Charter/Mandate, Devnet test vault and stable app identity.

03 Prove device-native G1

Physical wallet, exact transaction, RPC and before/after token states with consent and receipts.

04 Release-quality assurance

External usability, clean install, negative/recovery states and dApp Store path.

We do not infer future feature completeness, safety or adoption from the existence of a prototype.



The exception moves.

The boundary does not.

A more human experience of financial independence, governed by precise Solana authority.

SOURCE / github.com/Faadil1/cresco-key

REAL APK • REAL MWA • VERIFIED DEVNET RULES • KNOWN BLOCKERS

Inspect the receipts, not just our words.

Original onchain evidence. Developer Devnet signers; not a physical-wallet mobile payment.

5-unit in-bounds transfer	3PQLrxQCq3mkt6T14u9h5rUXD...xsBDZKfEaicS	INSPECT ↗
12-unit standing-limit refusal	BWaa6cAG3hjvvSeBCAAg2hpmt...DwbX3bVAZ8C2	INSPECT ↗
Guardian one-time allowance grant	24DRvkbxWvAiwLvn2zHNXa5Fj...J3xbUwWC682u	INSPECT ↗
Deployed distinct program	6SoGabSLX2YHMjFx1ynbz5nLF...RmszddU6DeJP	INSPECT ↗

Blocking case #76: allowance used flag reads false while current execute_payment_once returns AllowanceAlreadyUsed.

A fast, falsifiable walkthrough.

Evidence and instructions for an external evaluator without privileged accounts or secret keys.

00:00 / DISCOVER	Open APK; understand the boundary-first proposition.
00:15 / NO WALLET	Explore live Devnet Key read-only, no signature.
00:35 / MOBILE	Connect a compatible wallet, authorize and sign a proof message.
01:00 / ROLE	Explore Young/Guardian navigation and exact terms.
01:20 / PROOF	Open transaction and program links; see what was confirmed.
01:40 / HONESTY	Inspect issue #76 and the physical-wallet proof boundary.

APP: React Native / Expo · CONTRACT: Anchor / SPL Token · WALLET: Solana Mobile Wallet Adapter · NETWORK: Devnet